INCH-POUND

MIL-P-70456C (AR)
AMENDMENT 6
08 December 1993

SUPERSEDING
AMENDMENT 5
6 July 1993

MILITARY SPECIFICATION

PROPELLANT, STICK, M31A1E1
FOR USE IN CHARGE PROPELLING M203A1

This Amendment forms a part of Military Specification MIL-P-70456C (AR) dated 28 April 1989, and is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of Department of Defense.

PAGE 2

2.1.2 Add the following new drawing:
"9345098 - Body, Molded"

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3.4.3 Delete the number "29" and substitute with the number "28.5" wherever it appears in the paragraph.

3.9.1.2 Delete in its entirety and substitute new paragraph 3.9.1.2:

"3.9.1.2 145 degrees F pressure. At standard weapon conditions and recommended charge weight, the average pressure at 145 degrees F shall not exceed 56,000 pounds per square inch (psi) (see 4.4.2.5.1)."

PAGE 6

3.11 Delete in its entirety and substitute the following new paragraph:

"3.11 Blending process control. A propellant lot shall be a minimum of 130,000 lbs. unless otherwise specified by the contracting officer. The contractor shall assure that each lot has been sufficiently blended to meet the ballistic uniformity requirements (see 3.9). The contractor shall submit a blending process control document to the government specifying the blending process control steps which are considered critical for meeting the ballistic uniformity requirements. The blending process control document shall be submitted in accordance with 4.1.2."

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Add the following paragraphs:

"4.1.3 Nonconforming material. The contractor's quality program plan (see 6.2) shall include an effective and positive system for controlling nonconforming material to include mixes, batches, or sublots of propellant that do not meet compositional requirement (see 3.2 and 4.1.3.1). The system shall include procedures for nonconforming material identification, segregation, and disposition. All nonconforming supplies shall be positively identified to prevent unauthorized use, shipment and intermingling with conforming supplies.

4.1.3.1 Manufacturing process. The contractor's manufacturing process shall be documented as part of the quality program plan (see 6.2). Under a batch manufacturing process, the contractor shall maintain an effective and positive system (weighing each ingredient) for controlling amounts of ingredients charged to each mixer to assure the material meets the compositional requirements (see 3.2). Under a continuous process where the batch system is not applicable, the contractor shall maintain an effective and positive system for controlling amounts of ingredients charged to each mixer to assure the material meets the compositional requirements (see 3.2). For an unproven propellant manufacturing process which is significantly different from the present propellant manufacturing process, the contractor shall submit his proposed process control system for meeting compositional requirements in the form of a process control document to the contracting officer for approval by the technical agency. The contractor shall not initiate production without written approval of the process control document from the contracting officer. Once approved the contractor shall not change the process without approval from the contracting officer (see 6.13)."

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4.3.2.2 Delete in its entirety and substitute new paragraph 4.3.2.2:

"4.3.2.2 Testing to be performed. The following test program shall be used for the first article test. (The first article test is a qualification test for a new contractor or a major process change. Once a contractor has demonstrated the ability to produce acceptable stick propellants the first article test may be waived by the Contracting Officer. The

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propellant grains inhibited from each subsequent production lot shall

following test program (see Table III) is the minimum test requirement for the first article test and shall be used when the stick propellant manufacturing process is not significantly different from the present propellant manufacturer. If the new contractor's process or the proposed process change is found to be significantly different from the present manufacturing process, additional testing may be required to qualify the process difference.)

TABLE III. FIRST ARTICLE TEST.

Test Phase	Temp (deg F)	No. of Calibration Rds	No. of Test Rds	Test Meth Std
Propellant Checkout	70		10	TOP-4-2-504
	145		10	TOP-4-2-504
	-60		10	TOP-4-2-504
Sequential Environ-mental	145	15	60	TOP-4-2-504
	-60	15	60	TOP-4-2-504"

PAGE 11

4.4.2.5 Delete in its entirety and substitute new paragraph 4.4.2.5:

"4.4.2.5 Permissible individual maximum pressure. If at any time during testing, an individual pressure greater than that specified in 3.9.1.1 is observed, testing shall cease and the lot shall be rejected."

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4.4.2.5.1 Delete in its entirety and substitute new paragraph 4.4.2.5.1:

"4.4.2.5.1 Average 145 degrees F pressure. If the average pressure at 145 degrees F exceeds 56,000 pounds per square inch (psi), the lot shall be referred."

4.4.3 Delete the number "29" and substitute with the number "28.5".

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4.5.1.3 Delete in its entirety and substitute new paragraph 4.5.1.3:

"4.5.1.3 Length. Five hundred (500) sticks shall be randomly selected from each sublot and measured by gage for length. If one stick is found to be exceeding the maximum length requirement, the sublot shall be rejected. Sticks with a length shorter than the minimum requirement shall be classified by the degree of deviation from the minimum length. The basis for rejection of a sublot shall be: (1) Reject if more than 14 sticks are found shorter than 28-13/16 inches. (2) Reject if more than 7 sticks are found shorter than 28-9/16 inches. (3) Reject if one or more sticks are found shorter than 28-1/2 inches. A 100% inspection on stick length shall be done on the rejected sublot. Each stick shall be measured by gage and any stick which does not meet the length requirement should be removed from the sublot. The final blending on the entire lot should not be initiated until all sublots are found to be meeting the stick length requirement. The gage design for propellant length measurement shall be submitted to the technical agency for approval."

PAGE 14

4.5.1.4.1 Delete in its entirety and substitute new paragraph 4.5.1.4.1:

"4.5.1.4.1 Carbon black. The actual percentage of carbon black shall be determined in accordance with Method 308.1 or 309.1, MIL-STD-286."

4.5.1.5 Delete in its entirety and substitute new paragraph 4.5.1.5:

"4.5.1.5 Charge weight loadability. Five (5) 28.5 lb stick propellant samples shall be randomly withdrawn from the final blended lot. Each 28.5 lb stick propellant sample shall be weighed to 28.5 lb - 2.0 ounces. Each 28.5 lb stick propellant sample shall be bundled and tied as tight as possible with cords in four places (both ends and two places in the middle). An additive liner shall be wrapped around the stick bundle and tied as tight as possible with three cords (both ends and one place in the middle). The stick bundle with the additive liner shall be loaded into an inert cartridge case with dimensions as specified in Drawing #9345098. If any stick bundle with the additive liner cannot be loaded into the inert cartridge case without any difficulty, the propellant lot shall be rejected."

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4.5.3 Delete in its entirety and substitute the following new paragraph:

"4.5.3 Proving ground assessments (6.5) These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure using M549, RA or PXR-6325 projectiles inert loaded within 0.1 pound of an as fired weight of 96.0 pounds using primer, M82 and fired from M198 Howitzer with a M199 Cannon with 80 percent (minimum) remaining life. Firing shall be conducted with a spindle to charge base stand-off of 1 inch (plus or minus 1/4 inch). Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. One rear piezo gage (1) shall be used to measure the maximum chamber pressure and ignition delay of the charge fired at 145 degrees F and to measure the maximum chamber pressure and ignition delay of the charge fired at -65 degrees F (No piezo gage measurement is required for 70 degrees F firing). Velocity coils will be used to measure the muzzle velocity of each charge. Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply to all -65 deg F, 145 deg F and 70 deg F conditioning: -65 deg F from -65 deg F to -60 deg F; 145 deg F from 140 deg F to 145 deg F; 70 deg F from 67.5 deg F to 72.5 deg F. Temperature indicator (thermometer) strips shall be placed on the side (combustible cartridge case) of all charges conditioned at 145 deg F as a further assurance of the proper conditioning temperature. If any indicator strip shows the charge has been exposed to a temperature greater than 145 deg F, that charge will not be fired. Propelling charges used for test as calibration shall be assembled in accordance with MIL-C-70462 (see 6.9)."

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4.5.3.1.1 Add the following to the end of the para 4.5.3.1.1:

"The manufacturer of the combustible cases used in the calibration components for the propellant acceptance test shall be specified in the Propellant Acceptance Sheet along with the recommended charge weight (see 6.9.3)."

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* 4.5.3.2 Delete in its entirety and substitute the following:

"4.5.3.2 Uniformity. (see Table V)

4.5.3.2.1 Initial production. Thirty rounds shall be assembled with the calculated charge weight of propellant. Ten rounds each shall be conditioned to -65 deg F, 70 deg F and 145 deg F (as indicated by special thermocoupled round). Test rounds fired at 70 deg F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (see 3.9.1) and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The initial production lots represent the first three consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. After three accepted propellant lots from the same manufacturer, the uniformity acceptance test shall be fired in accordance with 4.5.3.2.2.

4.5.3.2.2 Subsequent production. Twenty rounds shall be assembled with the calculated charge weight of propellant (see 4.5.3.1) as determined for each lot for nine (9) consecutive lots. Ten rounds each shall be conditioned to 70 deg F and 145 deg F (as indicated by special thermocoupled round). Test rounds fired at 70 deg F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (see 3.9.1) and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. For every tenth propellant lot, ten additional rounds shall be assembled with the calculated charge weight of the propellant lot and conditioned to -65 deg F (as indicated by special thermocoupled round). Firing results at -65 deg F shall be corrected to standard projectile weight (see 3.9.1) and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The mean and standard deviation of the velocities and pressures at -65 deg F shall be calculated using corrected results. Throughout production, if there is a major change in manufacturing process, testing shall revert back to the initial production plan (see 4.5.3.2.1).

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6.2 Delete in its entirety and substitute new paragraph 6.2:

"6.2 Ordering data.

6.2.1 Contractor's quality program. Contracts should identify MIL-Q-9858 as the applicable higher level contract quality control requirement and specify data item description DI-R-5297A.

6.2.2 Data requirements. One copy of the Propellant Description Sheet and one copy of all ballistic test data for each lot of propellant shall be forwarded to ARDEC, Picatinny Arsenal, NJ 07806-5000, ATTN: SMCAR-QAR-R, and SMCAR-AEE-BP. Individual measurements of the total volatiles, stick weights, stick bundle weights and dimensions shall be recorded in the Propellant Description Sheet for proper evaluation of stick propellant."

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* 6.5 Add the following to TABLE V:

"**" next to -65.

"** Initial production only and every tenth propellant lot" at the bottom of the table.

PAGE 20

Add the following new paragraph:

"6.9.3 Recommended charge weight. The recommended charge weight as specified in the Propellant Acceptance Sheet may need final weight adjustment if the manufacturer of the combustible case components used for the propelling charge loading, assembling and packing is different from the manufacturer of the combustible case components used for calibration components which are used for the propellant acceptance test. The weight adjustment for the recommended charge weight shall be obtained from the first article test for the new contractor for manufacturing of the combustible case for the propelling charge as specified in MIL-C-70461."

NOTE: The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the

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